

fixed
 $\mathbf{u} = \mathbf{0}$

$L = 40 \text{ mm}$

Design domain Ω

$$\sigma_{\text{vm}}^{\text{app}} \leq \bar{\sigma} \eta(\bar{\rho})$$

$$\bar{\sigma} = 180 \text{ MPa}$$

Γ_t $l = 6 \text{ mm}$

Uniform tangential traction

$$\mathbf{t} = (0, -P/l)^T$$

$$P = 400 \text{ N}$$

$2L = 80 \text{ mm}$

$\boldsymbol{\sigma} \mathbf{n} = \mathbf{0}$ on the remaining boundary